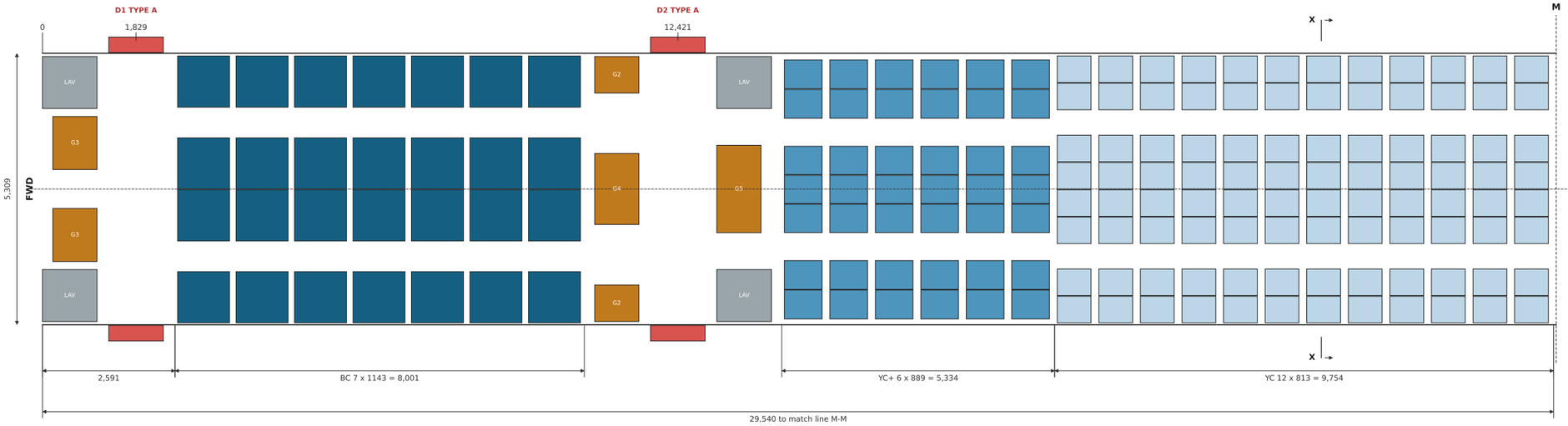
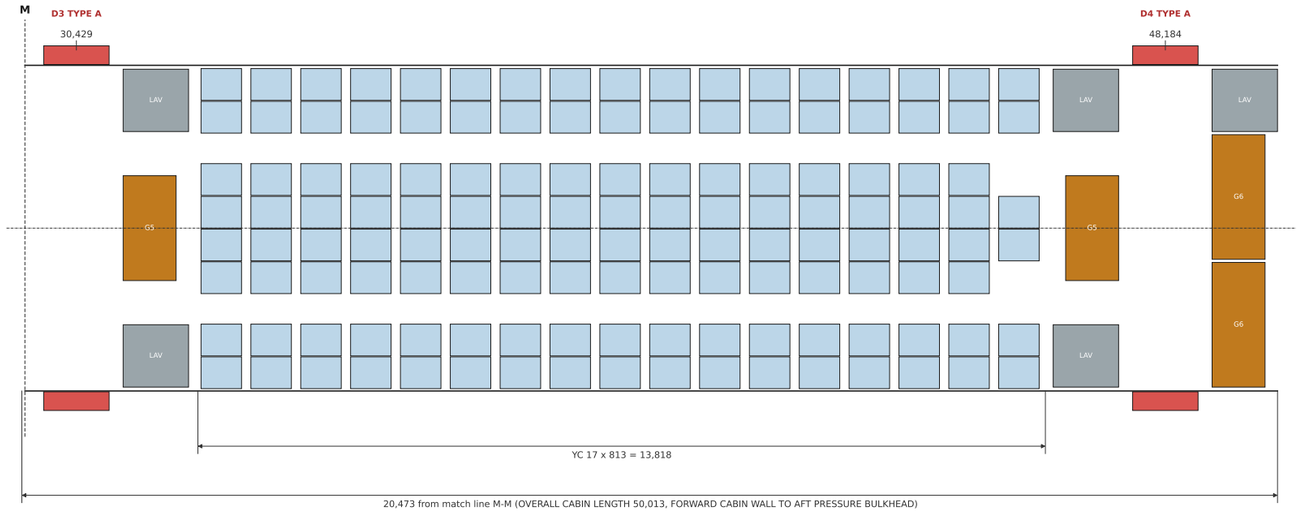
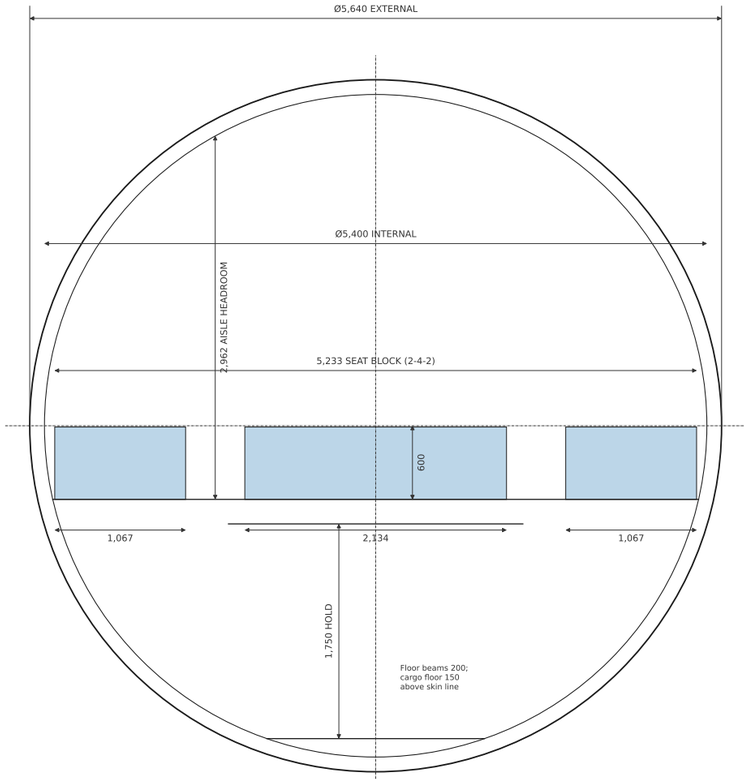



**VIEW A  PLAN, FORWARD CABIN (SCALE 1:100)**




**VIEW B  PLAN, AFT CABIN (SCALE 1:100), CONTINUED FROM MATCH LINE M-M**




**VIEW C  SECTION X-X, ECONOMY (SCALE 1:50)**

**LEGEND**

- Business (BC) 28 seats, 1-2-1, pitch 1143 (45 in), aisle 584 (23 in)
- Economy+ (YC+) 42 seats, 2-3-2, pitch 889 (35 in), aisle 533 (21 in)
- Economy (YC) 230 seats, 2-4-2, pitch 813 (32 in), aisle 483 (19 in)
- Galley, 864 (34 in) deep; Gn = n full trolleys (41 fitted, 40 required)
- Lavatory 1016 x 1067 (1680 in²), 9 total
- Type A exit pair, 1067 x 1829 clear (42 x 72 in), rating 110

**NOTES**

1. ALL DIMENSIONS IN MILLIMETRES UNLESS STATED. DO NOT SCALE. DOOR CENTRELINES DIMENSIONED FROM THE FORWARD CABIN WALL (DATUM 0).
2. LAYOUT TO TASK SPEC ME2618655 APP. 1 (3-CLASS MR RULES): SEAT RATIO, PITCH, SEAT/AISLE WIDTHS, TRAYS/TROLLEYS, LAVS, ATTENDANTS.
3. 300 SEATS: BC 28 (9.3 %), YC+ 42 (14.0 %), YC 230. CABIN CREW 11 (SERVICE 2 + 2 + 7; EXITS 4 x 2 = 8).
4. EXIT SPACING <= 18,290 (60 FT, CS 25.807): 10,592 / 18,009 / 17,755. 4 TYPE A PAIRS RATED 440 >= ~410 1-CLASS HD SEATS.
5. YC 4-SEAT CENTRE BLOCK TAKEN AS TWO DOUBLES (2 x 1067). WALKWAYS THROUGH GALLEY/LAV BANDS >= 660 (26 IN, APP. 1); COCKPIT PASSAGE 762.
6. WINDOW-SEAT WALL CLEARANCES: HEAD 178, SHOULDER 51, ARMREST 84, FOOT 108 (APP. 1 MIN 60 / 30 / 20 / 40).
7. BOX WING: FRONT WING BOX UNDER THE FLOOR BETWEEN D2 AND D3 (~40 % OF FUSELAGE LENGTH); REAR WING ON TWIN FINS ABOVE THE TAIL CONE.
8. FIRST CUT FOR CONCEPT EVALUATION. LD3 CONTOUR FIT IN THE HOLD TO BE CHECKED.

THIRD ANGLE

# BOX-WING CONCEPT: 3-CLASS MR CABIN LAYOUT

300-seat medium-range airliner on 100 % SAF, EIS 2040

**ENG3216 AEROSPACE GROUP DESIGN PROJECT, GROUP 10, UNIVERSITY OF SURREY**

| DRAWN | DATE | SCALE | SHEET |
|---|---|---|---|
| **N. Edwards** | **01/11/2026** | **1:100 / 1:50** | **1 OF 1** |
| DRG NO. | REV | UNITS | SIZE |
| **G10-PDR-CAB-001** | **A** | **mm** | **A3** |